Perimeter & Area

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: A rectangle is 9 cm long and 5 cm wide. What is its perimeter?
Method & answer: 2 × (9 + 5) = 2 × 14 = 28 cm.

Standard: A rectangle is 11 cm long and 6 cm wide. What is its area?
Method & answer: Area = 11 × 6 = 66 cm².

Advanced: A triangle has a base of 8 cm and a height of 5 cm. What is its area?
Method & answer: Area of a triangle = ½ × base × height = ½ × 8 × 5 = 20 cm².

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangle is 8 cm by 6 cm. What is its perimeter?

2. A square has sides of 9 cm. What is its perimeter?

3. A rectangle is 11 cm by 4 cm. What is its area?

4. Which shape has a perimeter of 30cm? A) a 9cm×6cm rectangle B) a square of side 7.5cm C) both A and B D) neither

5. A rectangle is 13 cm long and 4 cm wide. What is its perimeter?

6. A square has a perimeter of 28 cm. What is the length of one side?

7. What is the area of a 7cm by 7cm square? A) 14cm² B) 28cm² C) 42cm² D) 49cm²

8. A square has an area of 100 cm². What is the length of one side?

9. Select ALL of these shapes that have a perimeter of exactly 26cm: A) a 9cm×4cm rectangle B) a 10cm×3cm rectangle C) a 7cm×6cm rectangle D) a square of side 6.5cm E) an 8cm×5cm rectangle

10. A rectangle has a length of 12 cm and a width of 5 cm. What is its perimeter?

11. A rectangle is 14 cm long and 3 cm wide. What is its area?

12. Which rectangle has the larger area? A) 7cm×5cm B) 6cm×6cm C) they are equal D) cannot tell

13. A square has sides of 11 cm. What is its perimeter?

14. A rectangle is 8 cm by 8 cm. What shape is it, and what is its area?

15. A rectangle has an area of 56 cm² and a width of 7 cm. What is its length?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A triangle has a base of 6 cm and a height of 4 cm. What is its area?

2. Which of these has an area of 15cm²? A) a triangle with base 6cm, height 5cm B) a triangle with base 10cm, height 3cm C) both A and B D) neither

6. What is the area of a triangle with base 12cm and height 4cm? A) 16cm² B) 24cm² C) 48cm² D) 36cm²

7. A rectangular garden is 9 m long and 6 m wide. What is its area?

11. Two rectangles, 4cm×3cm and 3cm×2cm, are joined with no overlap. What is the total area?

12. A square lawn has a perimeter of 40 m. What is its area?

3. A triangle has a base of 10 cm and a height of 6 cm. What is its area?

4. An L-shape is a 7cm×4cm rectangle with a 2cm×2cm square cut from one corner. What is its area?

5. A rectangular field is 30 m by 18 m. What is its perimeter?

8. Select ALL of these triangles that have an area of MORE than 18cm²: A) base 8cm, height 5cm B) base 6cm, height 8cm C) base 10cm, height 3cm D) base 9cm, height 4cm E) base 5cm, height 4cm

9. A triangle has an area of 21 cm² and a base of 7 cm. What is its height?

10. A rectangle is 1.2 m long and 70 cm wide. What is its area in cm²?

13. A rectangle's length is twice its width. If the width is 5 cm, what is the perimeter?

14. A rectangle has a perimeter of 34 cm and a length of 10 cm. What is its area?

15. An L-shaped garden is a 10m×6m rectangle with a 3m×2m rectangle removed from a corner. What is its area?

Standard / Hard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. A rectangular field is 25 m by 12 m. A path 1 m wide is built all the way around the inside edge. What is the area of the path?

2. A rectangular lawn is 14 m by 9 m. A 4m×3m flower bed is cut from one corner. What is the remaining area of the lawn?

3. Which shape has the LARGEST area? A) a triangle base 10cm height 8cm B) a 6cm×7cm rectangle C) a 5cm×9cm rectangle D) a square of side 6cm

4. A triangle and a rectangle have equal areas. The rectangle is 9 cm by 4 cm, and the triangle has a base of 12 cm. What is the triangle's height?

5. A rectangular room measures 6 m by 4 m. Carpet costs £11 per m². What is the total cost of carpeting the room?

6. A square garden has an area of 144 m². What is the perimeter of the garden?

7. Select ALL of these statements that are TRUE about a triangle with base 14cm and height 6cm: A) its area is 84cm² B) its area is 42cm² C) it is exactly half of a 14cm×6cm rectangle D) its area cannot be found without more information

8. A 24cm×16cm piece of card has a 4cm×4cm square cut from each of its four corners. What is the remaining area?

9. A rectangle's length is 5 cm more than its width. If its perimeter is 30 cm, what are its dimensions?

10. A large rectangle 12m×8m has a 4m×3m rectangle removed from a corner, forming an L-shape. What is the perimeter of the L-shape?

11. A rectangular sports pitch is 60 m long and 40 m wide. A running track goes around the outside with a uniform width of 2 m. What is the perimeter of the outer edge of the track?

12. A square and a rectangle have equal areas. The square has sides of 10 cm and the rectangle's width is 5 cm. What is the rectangle's length?

13. An L-shaped room is made of two rectangles, 5m×4m and 2m×2m (no overlap). Carpet costs £13 per m². What is the total cost to carpet the room?

14. A rectangular banner's length and width are both doubled. If the original area was 20 cm², what is the new area?

15. A triangle has a base of 16 cm and a height that is half the base. What is its area?